Qiyan Jin

peterkim25162516-lab.github.io

peterkim25162516@gmail.com | China: +86 15844331419 | Singapore: +65 80319376



Mechanical Engineering graduate and NTU MSc student focused on robot manipulation learning, with experience in π0.5 training and evaluation, Unitree G1 teleoperation and cross-simulator testing of dexterous hands.

Education

Nanyang Technological University, Singapore
Aug 2026 - May 2027

School of Electrical and Electronic Engineering (EEE)

Master of Science (Computer Control & Automation) | Full-time; expected completion May 2027

Shanghai Jiao Tong University
Sep 2022 - Aug 2026

University of Michigan-Shanghai Jiao Tong University Joint Institute

Bachelor of Science in Engineering, Mechanical Engineering | Graduated | GPA: 3.0 / 4.0

Experience

DexMate | Robot Learning Intern | Singapore
Oct 2026 - Present

> **Project:** The team is using 3D Gaussian Splatting (3DGS) to reconstruct objects, robots and workbenches as visually realistic digital twins, with the goal of sim-to-real policy transfer.

- Designing a benchmark to compare manipulation policy performance in simulation and on real robots; currently defining test tasks and evaluation protocols.

Shanghai Institute for Physical Artificial Intelligence and Robotics (PAIR)
May 2026 - Oct 2026

Intern - Dexterous-Hand Simulation Benchmark (DexArena)

Paper status: Submitted to ICLR 2027

> **Project:** DexArena is a dexterous-manipulation simulation benchmark using everyday-object tasks to evaluate policies across hand models, scene variations and long-horizon tasks.

- Developed ManiSkill dexterous-hand tasks, refining observations, rewards and success checks across hand models. Adjusted Sharpa and Wuji simulated wrist cameras and checked hand and target-object visibility during manipulation.
- Designed and implemented full-parameter and LoRA-based supervised fine-tuning (SFT) pipelines for π0.5, adapting demonstration data and training models. Aligned joint order and state/action mappings for single- and dual-arm joint-position control.
- Evaluated both pipelines on 30 manipulation tasks, scene variations and long-horizon tasks. Adopted LoRA as the project's π0.5 baseline for its much lower training cost, at slightly lower overall success.
- Contributed manuscript methods and experimental setup, documenting model inputs, training, control and evaluation settings and checking them against experiment records.

Selected Projects

Unitree G1 Whole-Body VLA Simulation
May 2026 - Aug 2026

Undergraduate capstone | 5-person team

> **Project:** The team explored humanoid whole-body manipulation in MuJoCo, using teleoperated demonstrations to train GR00T N1.7 policies and SONIC WBC for whole-body control.

- Primarily handled PICO 4 / VR_3PT whole-body teleoperation. Helped calibrate the headset, controllers and simulated robot, and collected demonstrations of walking, grasping and bimanual handover.
- Reviewed task labels, episode boundaries and action completeness in long-horizon demonstrations; identified control anomalies and recollected low-quality trajectories for the team's training dataset.
- Reproduced control degradation when video recording was enabled, contributed to collection-software and workflow changes, and rechecked teleoperation with recording enabled.

Sharpa Wave Model Checks and Cross-Simulator Evaluation
Aug 2026 - Oct 2026

Independent project

Paper status: Independent manuscript, 2026

> **Project:** Compared motion, contact and object retention of publicly available Sharpa Wave hand models in MuJoCo and kit-less OVPhysX, studying how simulator settings affect evaluation.

- Built cross-simulator tests with aligned joints, scenes and control inputs; checked motion, fingertip trajectories and contact through repeated runs and timestep refinement.
- Stabilised contact readouts by changing stepping calls. Dry-friction controls explained most fingertip differences in the tested non-contact tasks; remaining causes were not fully isolated.
- Both backends retained a free ball; collision-disabled controls confirmed dependence on hand contact. Success did not imply matching trajectories, and some OVPhysX cases remained timestep-sensitive.

Technical Skills

Programming & Learning	Python, MATLAB, PyTorch; LeRobot, π0.5 supervised fine-tuning
Simulation & Evaluation	ManiSkill, MuJoCo, kit-less OVPhysX; task development, policy evaluation
Teleoperation & Data	Teleoperation calibration, demonstration collection and quality review
Design & Fabrication	SolidWorks / CAD, 3D printing, laser cutting, prototype assembly and testing

Additional Experience

Ronovo Surgical | Project Assistant Intern
Feb 2025 - Apr 2025

- Supported medical-robotics project scheduling and R&D milestone tracking, coordinating updates across hardware, software testing and supply chain teams.
- Organised technical documents and bills of materials, followed up on component procurement, and assisted with prototype assembly and basic testing.

Additional Engineering Projects

Dual-Arm Dental Navigation Simulation
Junior Fall

ME4670J Introduction to Robotics

- Derived a six-frame coordinate-transform chain between surgical and vision arms for an occluded target. Used Python and MATLAB to test inverse kinematics, configuration space and RRT* path planning.

Cylindrical Pier Inspection Climbing Robot
Junior Summer

ME3500J Design and Manufacturing II

- Managed scheduling and procurement; iterated a three-drive-unit chassis and spring-pin/one-way-slot self-locking linkage in CAD. Fabricated PLA and acrylic parts, then assembled and tested the prototype.

Automatic Card Dealer for Multiple Players
Sophomore Summer

ME2500J Design and Manufacturing I

- Designed the dealing mechanism and top-mounted electronics to rotate together, reducing cable winding. Adjusted the card-release lead for rotational speed to deal continuously to different positions.

Mobile Robot for Area Light-Intensity Mapping
Freshman Summer

ENGR1000J Introduction to Engineering

- Modelled the chassis in SolidWorks, helped debug the light sensor, and used Python to turn collected readings into an area light-intensity map.

Teaching and Campus Experience

Teaching Assistant | ME3950J Experiment (1)
Senior Fall

- Handled the Technical Communication component's formatting checks, helped grade assignments, and marked content that did not follow course requirements.

Teaching Assistant | ME2500J Design and Manufacturing I
Junior Spring & Summer

- Ran lab sessions and taught CAD across two course offerings. Guided mechanical design, component selection, basic machining and presentation preparation; helped with grading, grade entry and Canvas materials.

Academic Administration Assistant
Mar 2024 - Aug 2024

Undergraduate Education Office, UM-SJTU Joint Institute

- Supported course scheduling, TA coordination, course evaluation, examinations and grade administration, as well as course-selection guidance and degree audits.

Honours and Competitions

- First Prize, Shanghai Korean-Chinese University Student Scholarship | Mar 2024
- First Prize, ME2500J Game Day
- Participation Award, SJTU RoboMaster Campus Competition | Dec 2022
- Third Prize, SJTU "Liming Cup" Mechanical Creativity Competition | Apr 2023
- Outstanding Design Award, Summer Program Expo | 2022-2023 Academic Year